Wu 2020 Acute Myeloid Leukemia — analysis summary

RUN 3 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

iLISI 1.00 (fully patient-segregated) → 10.2 of 36 samples after Harmony (θ=10); kNN graph = 1 component. One continuous, uniformly low-cycling dome.

Gate 2 — module geography → GO (non-branching)

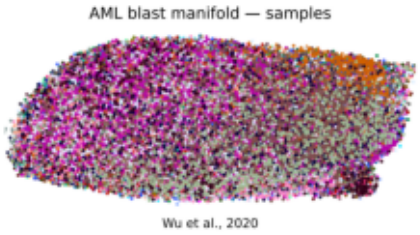
Organized Gast↔pSom cross-repression (r=−0.64) with a single terminal Crest gradient (Crest-high decile at pseudotime 0.86); no diverging arms.

Gate 3 — hub cross-repression → NA

Branching-only. Manifold is a fold-free curve (2 tips, 0 forks); no bifurcation to test.

STEP FIGURES

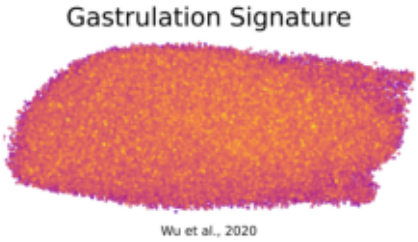
Manifold — samples



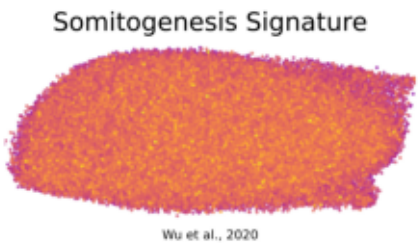
Pseudotime



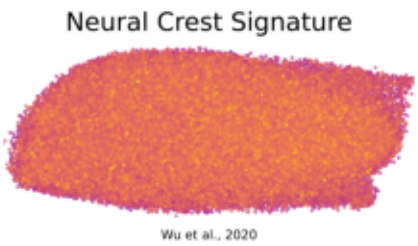
Gastrulation



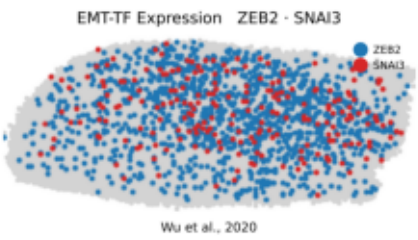
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Acute Myeloid Leukemia (Wu et al., 2020), Microwell-seq — 96,875 malignant cells (36/38 samples); counts (log1p of normalize_total).

Verdict.
Pattern reproduces (non-branching). Single continuous gastrulation → neural-crest axis with organized Gast↔pSom cross-repression; no bifurcation. Crest-arm inversion flag: none (Crest opposes Gast). Geometry: Linear. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs ZEB2 (5.1%) + SNAI3 (0.9%) — canonical EMT-TF panel largely silent in this myeloid malignancy.

Parameter choices

Min-cell threshold	≥100 cells (floor) → drop P18/P13; no dominant multiplicative cliff (top/second 1.35)	**Integration**	Harmony θ=10 (iLISI plateau; θ≥20 over-integrates); n_pcs=30, no sc.pp.scale
Geometry	scFates curve N=20, fold-free (2 tips/0 forks); PPT tree overfits 5–10 spurious branches	**Root**	Gast/pSom-high, Crest-low tip (origin; Crest late/terminal); cell-cycle-free axis
Cross-repression	Gast↔pSom r=−0.64; Crest-arm inversion flag: none (Crest opposes Gast)	**EMT-TFs**	ZEB2 (5.1%) + SNAI3 (0.9%) — canonical panel largely silent